

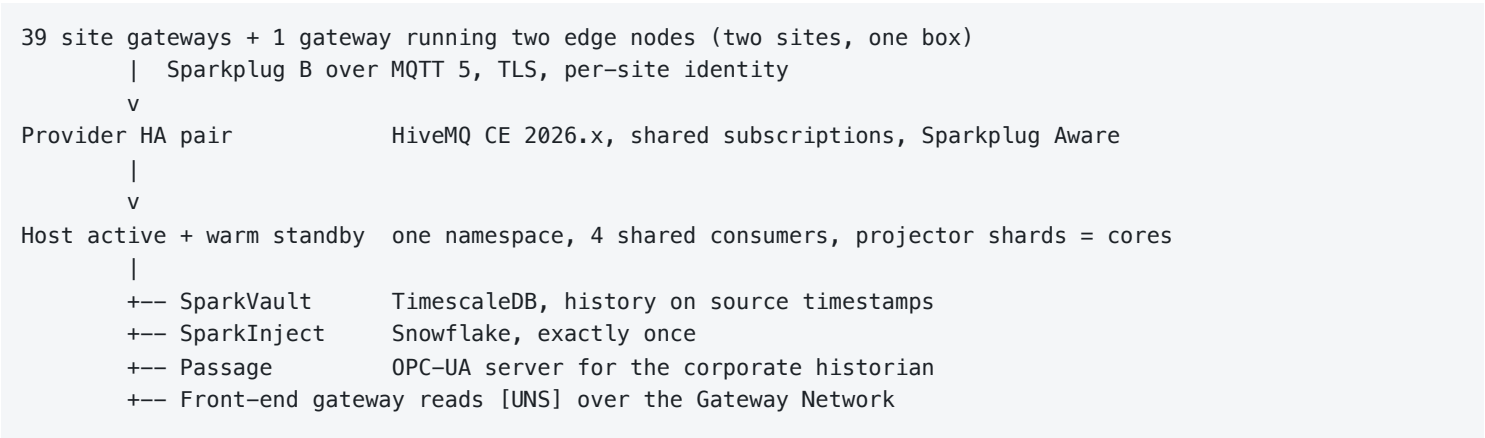
The 40-Site Ignition Reference Architecture

SparkBridge for Ignition. GreenGuard USA. September 2026. Every figure in this document names the machine it was measured on; the full record is at <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>.

1. The estate

Forty production sites, each running an Ignition gateway with 5,000 to 25,000 tags. Sites reach the center over mixed WAN: fiber at the plants, cellular at the remote pads, one satellite link. A front-end gateway serves several hundred Perspective users. Corporate wants history in one place, a warehouse feed, and OPC-UA access for a historian it already owns.

2. The shape



Three roles. Edge moves the data. Provider carries it. Host uses it.

3. Sizing

Layer	Sizing rule	Source
Edge per site	One module license. Up to 8 edge nodes per gateway. 250,000-tag budget per gateway	products page, Edge table
Wire metrics fleet-wide	40 sites x ~7,000 published leaves after metadata folding = ~280,000	migration plan calibration (Irving 8,700, Mesa 22,400, Dickinson 8,500 leaves)
Host memory	~300 bytes per wire metric per consumer; 500,000 wire metrics in under 4 GiB; 280k x 4 consumers ~ 340 MB session state plus the managed provider	PERFORMANCE.md section 7
Host CPU	Projector shards = core count. Six cores sustained at least 339,527 updates/s end-to-end on the Linux reference rig; four logical cores bend at 55,464	PERFORMANCE.md sections 0 and 8
Host recommendation	8 cores, 16 GB, 32 GB if SparkVault shares the box	sizing from the rows above
Provider	Two HiveMQ CE nodes, 4 cores, 8 GB each. No client-count license	docs/BROKER.md, docs/HA.md

Layer	Sizing rule	Source
Store-and-forward	50 MB default per edge node; at 2,697 messages/s durable and typical site change rates that is hours to days of outage. Size cellular and satellite sites at 500 MB	PERFORMANCE.md section 5

4. What happens when things fail

Failure	Behaviour	Measured
WAN drop at a site	Edge buffers to disk; Host marks the site's points bad, not current	2,697 msg/s durable
WAN returns	Reconnect within 10 s cap, rebirth, ordered replay on source timestamps with the historical flag; SparkVault backfills the curve	400,000 of 400,000 replayed; 40,035 upd/s on the six-core rig
Host primary crash	Standby promotes after the debounce; edges never rebirth	244 ms across 25 nodes, 5.7 ms mechanical
One Provider node down	Edges and Host walk to the other server in the set	Edge MultipleBroker TCK PASS, ~0.6 s server walk
Broker outage entirely	Every site buffers; nothing is lost up to buffer size	store-and-forward suite
A site's credential leaks	It can publish only its own group and node, never a command	Provider ACL, proven live

5. Adding consumers later

Each of these is a subscription at Host or a module on it. No change at the forty sites.

- Analytics or AI: subscribe to the shared stream, or read SparkVault, or take the Snowflake copy.
- A second corporate historian: Passage already serves OPC-UA; add a client.
- Estate-wide streaming past the tag ceiling: SparkFlow on Host.

6. Operations

- Site 41: install Edge, enter the enrollment code, \$995. Certificate issuance and automatic FleetOps registration are on the public roadmap.
- Fleet health: Host /fleet (buffering, headroom per edge), FleetOps (versions, drift, certificates).
- Security posture: SparkID generates the rules, Provider enforces them, Sentinel scores the result.
- Capacity: Host /capacity reports its ceiling from the shard count it runs.

7. List price

Item	Qty	Each	Total
SparkBridge Edge	39	\$995	\$38,805
SparkBridge Provider	2	\$1,995	\$3,990
SparkBridge Host	2	\$1,495	\$2,990

Item	Qty	Each	Total
SparkVault	1	\$3,995	\$3,995
Required total			\$49,780 once
Optional: SparkInject (Snowflake)	1	\$2,995	
Optional: Passage	1	\$1,995	
Optional: FleetOps	1	\$4,995	
Optional: Sentinel	1	\$4,995	
Support, optional		20% of list per year	\$9,956 on the required set

Nothing is metered. Development gateways are free and not counted.

8. Reproduce the numbers

```
./gradlew :common:loadTest --console=plain
./gradlew :common:loadTest --tests io.sparkbridge.perf.CapacityLoadTest
./gradlew :common:standbyDrill
```

Send us your topology for a review of your own estate: <https://mqtt.greenguard-usa.com/sparkbridge/architecture-review>